

Feature Engineering Assignment Summary

Task 1: What is a Feature?

A feature is an input variable used by a machine learning model to make predictions. Features represent measurable properties of the data. For example, in the hotel dataset, `lead_time` is a good feature because it directly influences cancellation probability, while `booking_id` would be a bad feature as it carries no predictive information.

Task 2: Curse of Dimensionality

As the number of features increases, the distance between data points becomes less meaningful. All points start appearing equally distant, making it harder for models to distinguish patterns. This is known as the curse of dimensionality. It highlights the importance of feature selection and engineering to reduce unnecessary dimensions.

Task 3: Numeric Preprocessing

Numeric preprocessing involves transforming numerical features to improve model performance. Binning groups continuous values into categories, binarization converts features into binary values, and scaling normalizes feature ranges. `RobustScaler` is often preferred when outliers are present, as it uses median and interquartile range instead of mean and standard deviation.

Task 4: Distance Metrics Impact

Distance-based algorithms like KNN rely heavily on feature scales. Without scaling, features with larger values dominate the distance calculation. Applying `StandardScaler` or `RobustScaler` ensures fair contribution of all features. Euclidean distance measures straight-line distance, while Manhattan distance sums absolute differences.

Task 5: End-to-End Pipeline

A pipeline automates preprocessing and model training steps. It ensures consistency and avoids data leakage. Using `ColumnTransformer`, different preprocessing steps can be applied to numeric and categorical features. Pipelines improve reproducibility and simplify cross-validation.

Task 6: Feature Extraction

Feature extraction creates new features from existing data. Examples include extracting month from dates, identifying weekends, and grouping customers. These features capture hidden patterns that improve predictive performance.

Task 7: Feature Construction

Feature construction involves creating new features such as ratios, interactions, and aggregated values. Examples include price per person, total nights, and interaction between price and lead time. Care must be taken to avoid data leakage by not using future or test data during construction.

Task 8: Feature Importance and Selection

Feature importance identifies which features contribute most to predictions. Methods include correlation, mutual information, and tree-based importance. Feature selection removes redundant or irrelevant features, improving model performance and interpretability.